

GravelKing Pro

JAX Authorship Integrity & Regeneration

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Purpose: document exactly how JAX lyric editing moved from ordinary AI response handling to a verifiable human-line preservation workflow.

Executive Summary

The new workflow distinguishes AI-generated lyric material from text directly edited or dictated by the creator. A creator can press **Regenerate (Keep Human Lines)** on an active JAX lyric card. The system snapshots the current lyric text, locks human-attributed ranges with SHA-256 hashes, asks JAX to regenerate only untouched AI lines, deterministically reassembles the result, and rejects the response if a locked line is not preserved verbatim.

Old behavior → new behavior

Area	Before	Now
Action surface	Read aloud and Save to song were available; regeneration had no dedicated guarded action.	A dedicated Regenerate (Keep Human Lines) button appears beside those actions.
Human attribution	Edits could be made, but regeneration had no card-level preservation lock.	The card is snapshotted and human-attributed ranges become locked records.
Generation instruction	A normal request could produce another AI lyric response without an enforced preservation contract.	JAX is instructed to regenerate only untouched AI lines and preserve locked lines verbatim.
Reassembly	The incoming response could become the new text.	The refreshed response is merged against the AI baseline and current edited text deterministically.
Verification	No dedicated post-regeneration preservation gate.	Every lock must remain verbatim or the response is rejected.
Certificate data	Certificate context was tied primarily to the draft/canvas flow.	The exact generator lyric text is hashed at render time and audit fields attach to the track/certificate record.

Exact workflow

- 1. Press.** The button is disabled while regeneration is active, preventing overlapping writes.
- 2. Snapshot.** The exact lyric text displayed in the active card is read before dispatch, with the immutable AI snapshot as comparison baseline.
- 3. Lock.** Human-attributed insertions receive text, offsets, **human: true**, and a SHA-256 hash. The current authorship percentage is logged before the request.
- 4. Constrain.** JAX receives the locked lines and instructions not to rewrite, paraphrase, move, or delete them.
- 5. Reassemble.** The refreshed AI lyrics are merged with deterministic line-position preservation, not fuzzy matching.
- 6. Verify.** Every locked line must appear verbatim. Failure leaves the existing card unchanged.
- 7. Commit.** The card, Song Generator field, draft block, AI snapshot, and authorship display update only after success.

Data and certificate fields

- **finalLyricsHash:** SHA-256 of the normalized final lyric text sent to the audio generation endpoint.
- **lyricsAuthorshipScore:** bounded 0–100 human authorship percentage.
- **lyricsAuthorshipLedger:** word/character-level attribution entries.
- **finalLyricsLabel:** **Certified Final Rendered Lyrics.**
- **aiSnapshot:** immutable AI output baseline retained per JAX response.

Important boundaries

- Prompt instructions alone do not earn human lyric credit; text must be applied through the lyric editor or preserved as a locked range.
- A regenerated AI response is a new AI snapshot; it does not rewrite history.
- The Song Generator field is the final rendering handoff and is hashed from its exact current contents.
- Certificate issuance still depends on the existing certification and copyright-screen gates.

Verification performed

Frontend and shared-authorship typechecks passed. The production frontend build passed. A focused preservation test changed surrounding AI lines while confirming the human-edited line remained verbatim. The web workflow was restarted after the change.

Implementation checklist

Area	Before	Now
UI	Dedicated button on active JAX lyric cards.	Complete
State lock	Human ranges snapshotted and hashed before dispatch.	Complete
Ledger	Current authorship percentage logged before regeneration.	Complete
Prompt constraint	Locked-line preservation instruction sent to JAX.	Complete
Reassembly	Deterministic human-line injection after response.	Complete
Verification	Locked lines survive 100% or response is rejected.	Complete
Render audit	Hash, score, ledger, certificate label persisted.	Complete
Regression check	Human line survives refreshed AI response.	Complete

Document note: This describes implemented application behavior and its verification boundary. It is not a legal opinion and does not itself certify authorship outside the server-issued certificate record.